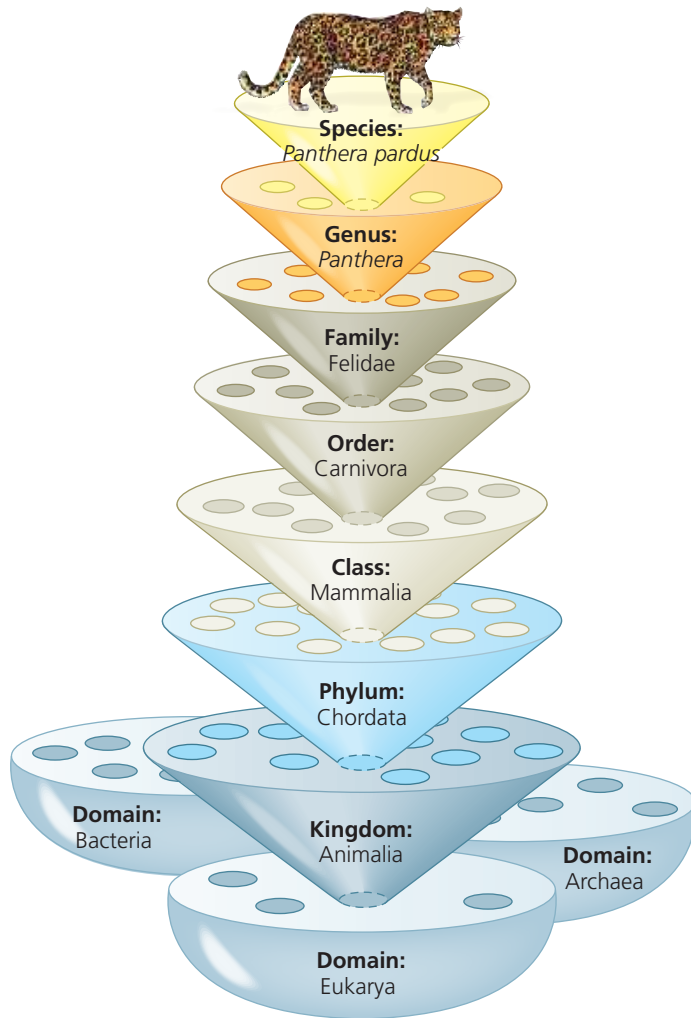


▼ **Figure 26.3 Linnaean classification.** At each level, or “rank,” species are placed in groups within more inclusive groups.

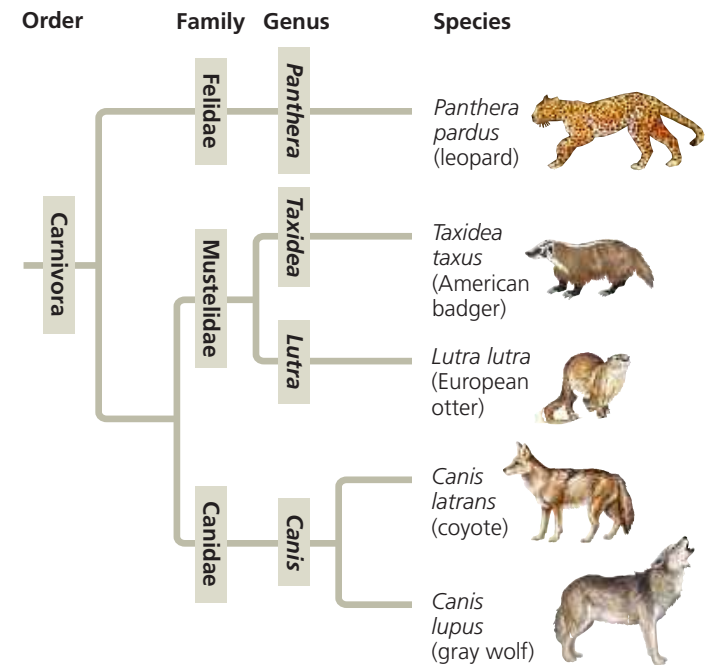


Linking Classification and Phylogeny

The evolutionary history of a group of organisms can be represented in a branching diagram called a **phylogenetic tree**. As in **Figure 26.4**, the branching pattern often matches how systematists have classified groups of organisms nested within more inclusive groups. Sometimes, however, systematists have placed a species within a genus (or other group) to which it is *not* most closely related. One reason for such a mistake might be that over the course of evolution, a species has lost a key feature shared by its close relatives. If DNA or other new evidence indicates that an organism has been misclassified, the organism may be reclassified to accurately reflect its evolutionary history. Another issue is that while the Linnaean system may distinguish groups, such as amphibians, mammals, reptiles, and other classes of vertebrates, it tells us nothing about these groups’ evolutionary relationships to one another.

Such difficulties in aligning Linnaean classification with phylogeny have led many systematists to propose that classification be based entirely on evolutionary relationships. In such systems, names are assigned only to groups that include a common

▼ **Figure 26.4 The connection between classification and phylogeny.** Hierarchical classification can reflect the branching patterns of phylogenetic trees. This tree shows evolutionary relationships between some of the taxa within order Carnivora, itself a branch of class Mammalia.



ancestor and all of its descendants. Using this approach, some commonly recognized groups become part of other groups previously at the same level of the Linnaean system. For example, because birds evolved from a group of reptiles, Aves (the Linnaean class to which birds are assigned) is considered a subgroup of Reptilia (also a class in the Linnaean system).

What We Can and Cannot Learn from Phylogenetic Trees

Regardless of how groups are named, a phylogenetic tree represents a hypothesis about evolutionary relationships. These relationships often are depicted as a series of dichotomies, or two-way branch points. Each **branch point** represents the common ancestor of the two evolutionary lineages diverging from it. An **evolutionary lineage** is a sequence of ancestral organisms leading to a particular descendant taxon. **Figure 26.5** illustrates these elements of a phylogenetic tree and offers more tips on interpreting these diagrams.

Each tree in Figure 26.5 has a branch point that represents the common ancestor of chimpanzees and humans. Chimps and humans are an example of **sister taxa**, groups of organisms that share an immediate common ancestor that is not shared by any other group. Note that there is a sister group associated with each branch point in a tree (since each branch point represents the common ancestor of the lineages diverging from it).

The members of a sister group are each other’s closest relatives, making sister groups a useful way to describe the